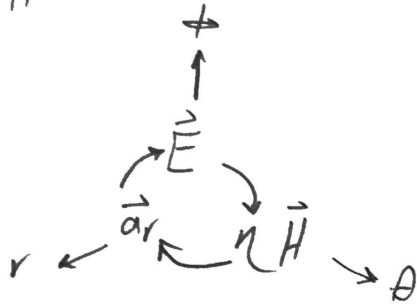


Q₁:

$$\vec{S} = \vec{E} \times \vec{H}$$



$$\eta \vec{H} = \vec{a}_r \times \vec{E}$$

$$\vec{H} = \frac{1}{\eta} [\vec{a}_r \times \vec{E}]$$

$$\vec{H} = \frac{1}{\eta} [\vec{a}_r \times \vec{E}]$$

$$\vec{H} = \frac{1}{2e^{j\frac{\pi}{6}}} [\vec{a}_r \times 2e^{-0.1r} e^{-jkr} \vec{a}_\phi]$$

$$\vec{H} = e^{-j\frac{\pi}{6}} e^{-0.1r} e^{-jkr} [\underbrace{\vec{a}_r \times \vec{a}_\phi}_{=-\vec{a}_\theta}]$$

$$\vec{H} = e^{-0.1r} e^{-jkr} e^{-j\frac{\pi}{6}} (-\vec{a}_\theta)$$

$$\vec{H} = -e^{-0.1r} \cos[\omega t - kr - \frac{\pi}{6}] \vec{a}_\theta$$

or

$$\vec{H} = e^{-0.1r} \cos[\omega t - kr + \frac{\pi}{3}] \vec{a}_\theta$$

[1]

$$|\eta| = \sqrt{(\sqrt{3})^2 + 1^2} = 2$$

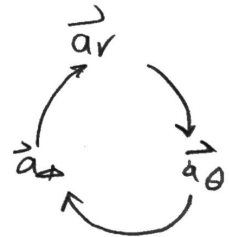
$$\theta_\eta = \tan^{-1}\left(\frac{\text{Im}}{\text{Re}}\right)$$

$$= \tan^{-1}\left(\frac{1}{\sqrt{3}}\right)$$

$$= 0.5235 \text{ rad}$$

$$= \frac{\pi}{6} \text{ rad}$$

$$\therefore \eta = \sqrt{3} + j = 2e^{j\frac{\pi}{6}} \Omega$$



$$\vec{a}_r \times \vec{a}_\phi = -\vec{a}_\theta$$

$$j = e^{j\frac{\pi}{2}}$$

$$-\frac{\pi}{6} + \frac{\pi}{2} = -\frac{\pi}{6} + \frac{3\pi}{6} = \frac{2\pi}{6} = \frac{\pi}{3} \text{ rad}$$

$$\begin{aligned}
 \therefore \vec{S} &= \vec{E} \times \vec{H} = 2e^{-0.1r} \cos[\omega t - kr] \vec{a}_\phi \times e^{-0.1r} \cos[\omega t - kr + \frac{\pi}{3}] \vec{a}_\theta \\
 &= 2e^{-0.2r} \cos[\omega t - kr] \cos[\omega t - kr + \frac{\pi}{3}] \underbrace{[\vec{a}_\phi \times \vec{a}_\theta]}_{-\vec{a}_r} \\
 &= -2e^{-0.2r} \cos[\omega t - kr] \cos[\omega t - kr + \frac{\pi}{3}] \vec{a}_r \quad \frac{W}{m^2}
 \end{aligned}$$

$$\begin{aligned}
 \therefore \vec{S} &= \vec{E} \times \vec{H} = 2e^{-0.1r} \cos[\omega t - kr] \vec{a}_\phi \times e^{-0.1r} \cos[\omega t - kr + \frac{\pi}{3}] \vec{a}_\theta \\
 &= 2e^{-0.2r} \cos[\omega t - kr] \cos[\omega t - kr + \frac{\pi}{3}] \underbrace{[\vec{a}_\phi \times \vec{a}_\theta]}_{-\vec{a}_r} \\
 &= -2e^{-0.2r} \cos[\omega t - kr] \cos[\omega t - kr + \frac{\pi}{3}] \vec{a}_r \quad \frac{W}{m^2}
 \end{aligned}$$

Q2:

The received average power density at Antenna (b) is

$$\vec{S}_{\text{ave}} = \frac{|\vec{E}_{\text{max}}|^2}{2|\eta|} e^{-\alpha z} \cos \theta_{\eta}$$

Medium (3)

$$\sigma = 4 \text{ S/m} \Rightarrow \frac{\sigma}{\omega \epsilon} = \frac{4}{(40\pi)(75 \times \left[\frac{10^9}{36\pi}\right])} = 48 \times 10^6 > 100$$

$$\epsilon_r = 75$$

$$\mu_r = 1$$

$\therefore$ Good-Conductor medium

$$\therefore \alpha = \sqrt{\pi f \sigma \mu} = \sqrt{\pi (20)(4)(4\pi \times 10^{-7})} = 0.0177 \quad \frac{\text{NP}}{\text{m}} = \alpha_3$$

$$\eta = (1+j) \frac{\alpha_3}{\sigma} = (1+j) 4.442 \times 10^{-3} \quad \Omega$$

$$|\eta| = 6.283 \times 10^{-3} \quad \Omega$$

$$\theta_{\eta} = \tan^{-1} \left(\frac{\text{Im}}{\text{Re}} \right) = \frac{\pi}{4} \text{ rad}$$

$$\vec{E}_{\text{max}} \Big|_{z=100} = \vec{E}_{\text{max}} \Big|_{z=0.03} e^{-\alpha_3 z} = \vec{E}_{\text{max}} \Big|_{z=0.03} e^{-0.03 \alpha_3} \quad \text{--- (1)}$$

but,

$$\vec{E}_{\text{max}} \Big|_{z=0.03} = \vec{E}_{\text{max}} \Big|_{z=1} e^{-\alpha_1 z} = \vec{E}_{\text{max}} \Big|_{z=1} e^{-\alpha_1} \quad \text{--- (2)}$$

$$\therefore \left| \vec{E}_{\max} \right|_{z=100} = \left(\left| \vec{E}_{\max} \right|_{z=1} e^{-\alpha_1} \right) e^{-0.03\alpha_2} = 100 e^{-[\alpha_1 + 0.03\alpha_2]} \vec{a}_y$$

$$\alpha_1 \text{ is at medium (1)} \Rightarrow \frac{\hat{\alpha}}{W\epsilon} = 0$$

$\therefore$ loss-less medium

$$\alpha_1 = 0$$

$$\alpha_2 \text{ is at medium (2)} \Rightarrow \frac{\hat{\alpha}}{W\epsilon} = \frac{10^5}{(40\pi) \left(\frac{10^{-9}}{36\pi} \right)} = 9 \times 10^{13} > 100$$

$\therefore$ Good-Conductor medium

$$\alpha_2 = \sqrt{\pi f \mu \sigma} = \sqrt{(9)(20)(4000 \times 4\pi \times 10^{-7})(10^9)} = 177.715 \text{ Np/m}$$

$$\therefore \left| \vec{E}_{\max} \right|_{z=100} = 100 e^{-[0 + 0.03(177.715)]} \vec{a}_y = 0.483 \vec{a}_y$$

$$\left| \left| \vec{E}_{\max} \right|_{z=100} \right|^2 = 0.233$$

$$\vec{S}_{ave} = \frac{(0.233)}{2(6.283 \times 10^{-3})} e^{-2(0.0177)(100)} \quad \text{Gs} \quad \frac{\pi}{4} \quad \vec{a}_z$$

$$\vec{S}_{ave} = 0.381 \vec{a}_z \quad \frac{W}{m^2}$$

